

MACHINE DESIGN

III B. Tech. I Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME23	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

Machine Design focus mainly on design of power transmitting elements like gears, Connecting rod, crankpin, crankshafts, pistons, cylinders, bearings, belts, ropes, chain's, pulleys, Power Screws and nuts.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Identify design variables and performance factors in the study of journal bearings
2. Apply the basic principles for design of piston, connecting rod and crankshaft based on maximum bending and twisting moments
3. Design screws that are applied in hosting of heavy weights like screw jack etc.
4. Explain power transmission elements like belts, chains and ropes and able to design on stress criteria.
5. Apply the design concepts to evaluate the strength of the gear.

UNIT-I	BEARINGS
Types of Journal bearings – lubrication – Bearing modulus – Full and partial bearings, Partial bearings – Clearance ratio – Heat dissipation of bearings, bearing materials – journal bearing design – Ball and roller bearings – Static loading of ball and roller bearings, bearing life.	

UNIT-II	ENGINE PARTS
Connecting rod: Thrust in connecting rod – stress due to whipping action on connecting rod ends – Cranks and Crank shafts, strength and proportions of over hung and center cranks – Crank pins, Crank shafts. Pistons, Forces acting on piston – Construction Design and proportions of piston, Cylinder, Cylinder liners.	

UNIT-III	POWER SCREWS
Types of screw threads for power screws, Torque required to raise and lower the loads by square threaded screws, Efficiency of square threaded screws, Overhauling and self-locking, ACME threads, Stresses in power screws, Design of screw jack.	

UNIT-IV	BELTS & PULLEYS
Transmission of power by Belt and Rope drives, Transmission efficiencies, Belts – Flat and V types – Ropes – Pulleys for belt and rope drives, Materials, Chain drives.	

UNIT-V	GEARS
Spur gears – Helical gears – Load concentration factor – Dynamic load factor, Surface compressive strength, Bending strength – Design analysis of spur gears – Estimation of centre distance, module and face width, check for plastic deformation, check for dynamic and wear considerations.	

Text Books:

1. V. Bandari, A Text Book of Design of Machine Elements, 3rd Edition, Tata McGraw Hill Education (P) Ltd., New Delhi, India, 2011.
2. R. L. Norton, Machine Design (An Integrated approach), 2nd Edition, Pearson Publishers, Chennai, India, 2006.

Reference Books:

1. R. S. Khurmi, A Text book of Machine Design, 14th Edition, S. Chand & Company, New Delhi, India, 2018.
2. S. MD. Jalaludin,(2011), Machine Design, 3rd Edition, Anuradha Publishers, Chennai, India, 2011.